Flashcard 1: What is Virtual Memory?

- **Definition:** A memory management technique that provides the illusion of having more memory than physically available.
- **Key Features:**
    - Allows large programs to run with limited physical memory.
    - Provides memory isolation between processes.
    - Enables multitasking by efficiently managing memory.
- **Example:** Physical RAM = 4 GB, Virtual memory extends to 16 GB using 12 GB of disk space.

Flashcard 2: What is Demand Paging?

- **Definition:** A technique where pages are loaded into memory only when required.
- **How It Works:**
    1. Process starts execution.
    2. Page fault occurs if the required page is not in memory.
    3. OS fetches the page from secondary storage.
- **Advantages:**
    1. Reduces initial loading time.
    2. Saves memory by loading only necessary pages.
- **Example:** A program requires 10 MB of memory but only 2 MB of pages are active at a time.

Flashcard 3: What is a Page Fault?

- **Definition:** Occurs when a program tries to access a page not currently in RAM.
- **Steps During a Page Fault:**
    1. OS pauses the process.
    2. Identifies the missing page.
    3. Loads the page from secondary storage into RAM.
    4. Updates the page table.
    5. Resumes the process.
- **Types:**
    1. Major Page Fault: Page not in RAM, loaded from disk.
    2. Minor Page Fault: Page in memory but not mapped to the process's page table.

Flashcard 4: What are Page Replacement Algorithms?

- **Definition:** Algorithms used to decide which page to replace when physical memory is full.
- **Types of Algorithms:**
    - **FIFO (First In, First Out):** Replaces the oldest page.
    - **LRU (Least Recently Used):** Replaces the least recently used page.
    - **Optimal Page Replacement:** Replaces the page that will not be used for the longest time in the future.

    - **Clock Algorithm:** Uses a reference bit to decide replacement.

Flashcard 5: FIFO Page Replacement Algorithm

- **Definition:** Replaces the oldest page in memory.
- **Advantages:** Simple to implement.
- **Disadvantages:** May replace frequently used pages.
- **Example:** Pages: [1, 2, 3, 4], Access Sequence: 1 → 2 → 3 → 4 → 5, Replacement: Page 1 is replaced first.

Flashcard 6: LRU Page Replacement Algorithm

- **Definition:** Replaces the page that has not been used for the longest time.
- **Advantages:** Reduces the chance of replacing frequently used pages.
- **Disadvantages:** Requires tracking access times.
- **Example:** Pages: [1, 2, 3], Access Sequence: 1 → 2 → 3 → 1 → 4, Replacement: Page 2 is replaced.

Flashcard 7: Optimal Page Replacement Algorithm

- **Definition:** Replaces the page that will not be used for the longest time in the future.
- **Advantages:** Theoretically the best algorithm.
- **Disadvantages:** Requires future knowledge of page references.
- **Example:** Pages: [1, 2, 3, 4], Access Sequence: 1 → 2 → 3 → 1 → 4 → 2 → 3, Replacement: Page 4 is replaced.

Flashcard 8: Clock Algorithm

- **Definition:** A practical approximation of LRU using a "clock" mechanism.
- **How It Works:** Uses a reference bit to decide replacement.
- **Advantages:** Simple and efficient.
- **Example:** Pages with reference bits: [1(1), 2(0), 3(0)], Page with reference bit 0 is replaced.

Flashcard 9: Illustrative Example of Page Replacement Algorithms

- **Problem:** Physical memory: 3 frames, Page Reference Sequence: [7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3].
- **Results:**
    - FIFO: 9 Page Faults
    - LRU: 8 Page Faults
    - Optimal: 6 Page Faults

These flashcards can be used for effective studying and quick revision of the concepts related to virtual memory and its management techniques.